THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Weight is measured in: 1

(a) kilograms **(b)** litres **(c)** newtons **(d)** grams

2. On which listed body is the weight of 1 kg the greatest? 1

(a) Jupiter **(b)** Moon **(c)** Venus **(d)** Mars

3. In a spring balance, 2 N is divided into 10 equal parts. Each division is: 1

(a) 2 N **(b)** 0.1 N **(c)** 0.5 N **(d)** 0.2 N

4. Archimedes' principle states that the upthrust equals the: 1

(a) mass of object **(b)** volume of container

(c) temperature of liquid **(d)** weight of liquid displaced

5. Friction also acts on objects moving through: 1

(a) vacuum **(b)** only solids

(c) liquids and gases **(d)** no medium

6. Blowing between two balloons makes them: 1

(a) burst **(b)** move apart

(c) move towards each other **(d)** fall down

7. Thunder is the loud sound produced when: 1

(a) rain falls **(b)** lightning heats and expands air

(c) wind stops **(d)** clouds collide like solids

8. A lightning conductor is a: 1

(a) glass rod

(b) wooden pole

(c) metal rod that carries charge to the ground

(d) rubber rod

9. The wall of seawater pushed by cyclone winds is called: 1

(a) flood plain **(b)** tide **(c)** storm surge **(d)** tsunami

10. Bordoila is a local storm of: 1

(a) Sikkim **(b)** Gujarat **(c)** Goa **(d)** Assam

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. When does an object float? When does it sink? 2

12. Why does a mug feel lighter inside water? 2

13. What is the eye of a cyclone? 2

14. Write two safety measures during lightning. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why is it easier to slip on a wet floor? 3

16. A block of weight 60 N has a base area of 0.2 m² on the floor. Find the pressure on the floor. 3

17. State any three differences between mass and weight. 3

OR

Draw a labelled diagram of a spring balance and name its parts. 3

18. Why do people carrying loads on head use a round cloth? 3

OR

Give three observations that show high-speed winds have low pressure. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain how frictional force can be shown with a block on rough and smooth surfaces and give your prediction. 5

OR

Describe how forces act when a swing moves to its highest point and comes back, naming the forces and their effects. 5

20. Explain why a rubber sucker sticks and why it becomes hard to remove, using the idea of pressure differences. 5

OR

Write an account of how air pressure and wind are connected, including breeze, storm and cyclone. 5

END OF QUESTION PAPER — Total: 40 Marks

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